

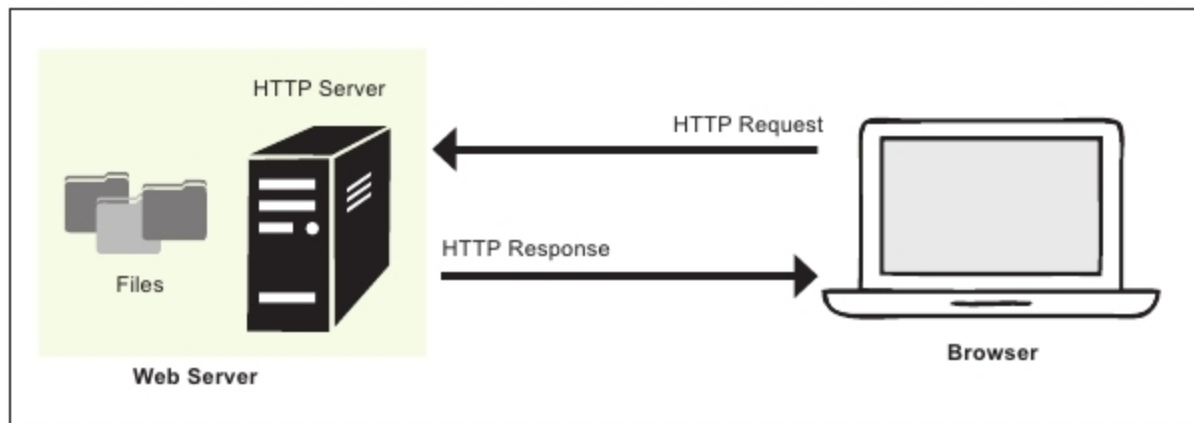


Figure 1: How the Web server works

This article lists out the top ten open source Web browsers in 2019.

Apache HTTP

Apache HTTP is an open source, cross-platform Web server software, released under the terms of Apache License 2.0. It is developed and maintained by an open community of developers under the Apache Software Foundation. It powers around 46 per cent of the world's websites. It works on a large number of operating systems, including Linux, FreeBSD, Solaris, Windows, MacOS X, Novell Netware, OS/2, and so on (and probably any UNIX-like system).

Apache supports a variety of features, many implemented as compiled modules which extend the core functionality. These can range from authentication schemes to supporting server-side programming languages such as Perl, Python, Tcl and PHP. Popular authentication modules include *mod_access*, *mod_auth*, *mod_digest*, and *mod_auth_digest*, the successor to *mod_digest*. A sample of other features include Secure Sockets Layer and Transport Layer Security support (*mod_ssl*), a proxy module (*mod_proxy*), a URL rewriting module (*mod_rewrite*), custom log files (*mod_log_config*), and filtering support (*mod_include* and *mod_ext_filter*). Popular compression methods on Apache include the external extension module, *mod_gzip*, implemented to help with reducing the size (weight) of Web pages served over HTTP. ModSecurity is an open source intrusion detection and prevention engine for Web applications.

Apache logs can be analysed through a Web browser using free scripts, such as AWStats/W3Perl or Visitors.

Virtual hosting allows one Apache installation to serve many different websites. For example, one computer with one Apache installation could simultaneously serve *example.com*, *example.org*, *test47.test-server.example.edu*, etc.

Features

- Loading of dynamic modules, handling of static files, index files, auto-indexing and content negotiation.
- IPv6 and HTTP/2 support.
- Custom logging and rotation, IP address-based geo location, user and session tracking, concurrent connection limiting, CGI support, FTP and XML support.
- URL rewriting, fine-grained authentication and authorisation access control.
- Fault tolerance and failover handling, and multiple load balancing mechanisms.

Latest version: 2.4.41

Official website: <https://httpd.apache.org/>

Apache Tomcat

Apache Tomcat, an open source Web server written in Java, was designed by the Apache Foundation with the credit of invention given to James Duncan Davidson, a software architect. It is termed an open source implementation of the Java-Servlet, JavaServer

Pages, Java Expression Language and WebSocket technologies. Apache Tomcat software powers numerous large-scale, mission-critical Web applications across a diverse range of industries and organisations.

Tomcat is made up of multiple components — Catalina, Coyote, Jasper, Cluster, etc. Catalina is termed as Tomcat's servlet container. Coyote is the connector component for Tomcat that supports HTTP 1.1. as the Web server, allowing Catalina, nominally a Java servlet or JSP container, to also act as a plain Web server that serves local files as HTTP documents. Jasper is Tomcat's JSP engine. It parses JSP files to compile them into Java code as servlets that can be handled by Catalina. The Cluster component is added to manage large applications. It is used for load balancing, which can be achieved through many techniques. Clustering support currently requires the JDK version 1.5 or higher.

Features

- Lightweight and highly flexible.
- Very stable and has an extra level of security.
- Well-documented and the most widely used Java application server.
- Supports HTTP/2, OpenSSL for TLS support with the JSSE connectors.
- Support for TLS virtual hosting, Web application memory leak protection and detection.
- Advanced IO capabilities and refactored clustering.

Latest version: 9.0.30

Official website: <https://tomcat.apache.org/>

Nginx

Nginx is a free and open source Web server based on a BSD-like licence. It is also an HTTP and reverse proxy server, mail proxy server and a generic TCP/UDP proxy server. It has been designed by Igor Sysoev. Nginx is capable of handling more than 10,000